

Detect a Phishing URL Using Machine Learning in Python



In a phishing attack, a user is sent a mail or a message that has a misleading URL, using which the attacker can collect important data like the passwords of the banks your money is in. This article gives a short tutorial on how to detect such phishing attempts.

Training the machine to predict

Once we have the data sets, we need to import the required libraries using the following lines of code:

```
import pandas as pd
from sklearn.ensemble import
RandomForestClassifier
```

If you do not have the libraries, you can use *pip* to install the libraries, as shown in Figure 1.

Once this is done, you can import the data sets and convert them into *pandas dataframe* for further processing using the following lines of code:

```
legitimate_urls = pd.read_csv("/home/
jishnusaarav/jupyter/Phishing-Website-
Detection/datasets/legitimate-urls.
csv")
phishing_urls = pd.read_csv("/home/
jishnusaarav/jupyter/Phishing-Website-
Detection/datasets/phishing-urls.csv")
```

After successful import, we need to merge both the dataframes — the

Through phishing attacks, attackers acquire important credentials that can be used for getting access to your bank or other financial accounts. The URLs sent by the attacker look exactly the same as the original applications we use on a daily basis. That is why people often believe these and enter their personal details. A phishing URL can open a Web page that looks similar to the original login page of your bank. Detecting such URLs has become very important of late as such phishing attacks are becoming pretty common. So let's see how we can check whether a URL is a misleading one or a genuine one using machine learning in Python, as it can help us see the code as well as

the outputs. We will be using Jupyter Notebook. You can use Google Colab or Amazon Sagemaker too, if you are more comfortable with those.

Download the data sets

To start, we will need the data set to work upon. You can download the data sets from the links given below.

Genuine URLs: <https://github.com/jishnusaarav/Phishing-attack-PCAP-analysis-using-scapy/blob/master/Phishing-Website-Detection/datasets/legitimate-urls.csv>

Phishing URLs: <https://github.com/jishnusaarav/Phishing-attack-PCAP-analysis-using-scapy/blob/master/Phishing-Website-Detection/datasets/phishing-urls.csv>